EXPERIMENT NO: 07

TITLE: Develop a simple calculator using XMLRPC

1. AIM

To design, implement, and verify a remote calculator application using the XML-RPC protocol in Python based on client-server architecture.

2. PREREQUISITES

- Basic understanding of Client-Server architecture and Socket communication. •

  Knowledge of Python input/output operations, function definitions, and data types.

- Fundamental concepts of standard application layer protocols (HTTP) and serialization formats (XML).

3. ALGORITHM / SYSTEM PROCEDURE

Server Program Procedure (calciserver.py):

1. Import SimpleXMLRPCServer from the standard Python xmlrpc.server module.
2. Define procedural functions for basic arithmetic operations: add, subtract, multiply, and divide. Include a logical check in divide to prevent division by zero.
3. Instantiate the SimpleXMLRPCServer bound to hostname "localhost" and port 8000.
4. Register the procedural functions with the server using server.register_function().
5. Execute server.serve_forever() to enter an infinite loop waiting for incoming RPC client calls.

Client Program Procedure (calciclient.py):

1. Import xmlrpc.client.
2. Connect to the server running at "http://localhost:8000/" using xmlrpc.client.ServerProxy().
3. Prompt the user for two floating-point numbers (a and b) using input().
4. Invoke the remote methods (proxy.add, proxy.subtract, proxy.multiply, proxy.divide) on the proxy object.
5. Print the output returned by the server.

4. PROGRAM CODE

Server Code: calciserver.py

```
from xmlrpc.server import
SimpleXMLRPCServer def add(a, b):
 return a + b
def subtract(a, b):
 return a - b
def multiply(a, b):
 return a * b
def divide(a, b):
 if b == 0:
 return "Division by zero not allowed"
 return a / b

server = SimpleXMLRPCServer(("localhost", 8000))
print("XML-RPC Calculator Server Running on Port
8000...")

server.register_function(add, "add")
server.register_function(subtract, "subtract")
server.register_function(multiply, "multiply")
server.register_function(divide, "divide")

server.serve_forever()
```

Client Code: calciclient.py

```
import xmlrpc.client

proxy = xmlrpc.client.ServerProxy("http://localhost:8000/")

a = float(input("Enter First Number: "))
b = float(input("Enter Second Number: "))

print("\nAddition =", proxy.add(a, b))
print("Subtraction =", proxy.subtract(a, b))
```

```
print("Multiplication =", proxy.multiply(a, b))
print("Division =", proxy.divide(a, b))
```

5. EXECUTION STEPS & SAMPLE OUTPUT

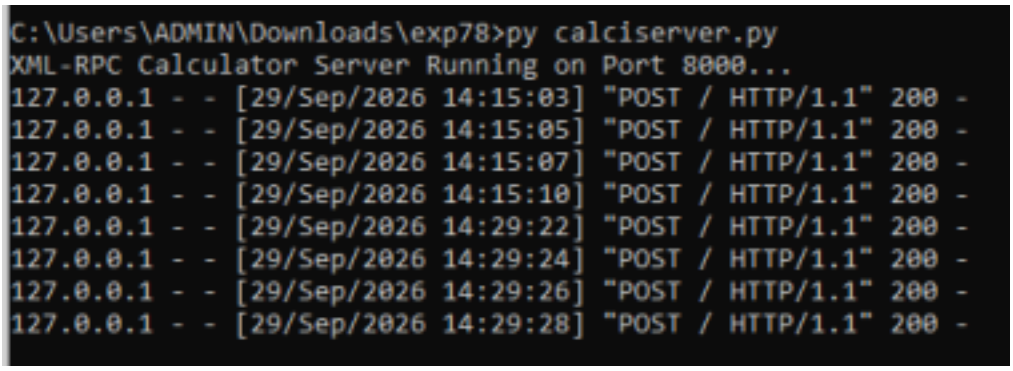
Step 1: Start the Server Process

Open a terminal / Command Prompt window and run:

Bash

python calciserver.py

Server Terminal Output:



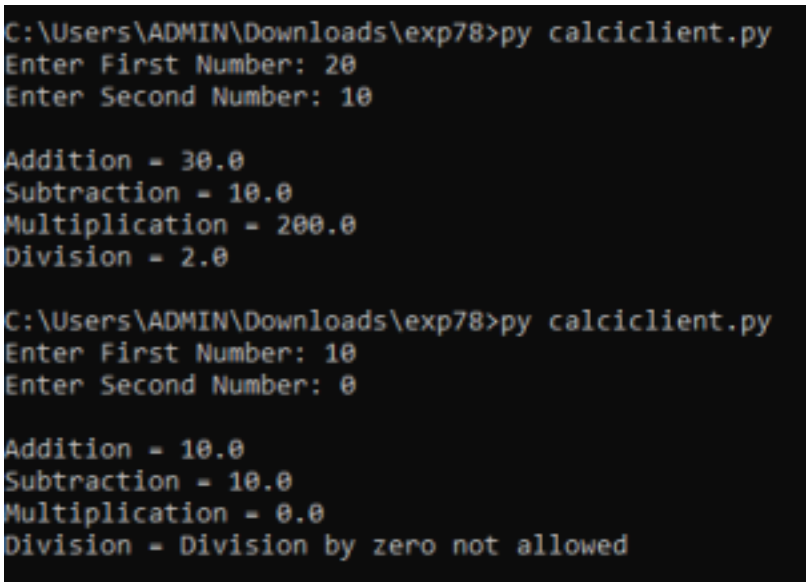
Step 2: Run the Client Process

Open a terminal / Command Prompt window and run:

Bash

python calciclient.py

Client Terminal Output (Standard Operation):



6. RESULT

The XML-RPC based simple calculator application was successfully executed. The procedures for addition, subtraction, multiplication, and division were registered on the server and remotely invoked by the client application.